IGCSE 0607 Practice Questions — E1.16 — Combined Questions & Markscheme

Exponential growth and decay

Total Questions: 9 — **Total Marks:** 50 Marks

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1 A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

(a) **(a)** Work out the value of p.

$p =$ ..... **[3]**

(b) Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

2 A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

3 A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

4 The population of a town is 65000.
The population increases exponentially at a rate of 2.8% per year.

(a) Calculate the expected population of the town after 5 years.
Give your answer correct to the nearest integer.

..... **[2]**

(b) Calculate the overall percentage increase in the population over the 5 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

5 The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

(a) **(a)** Calculate the value of r, correct to 3 significant figures.

$r =$ ..... **[3]**

(b) Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

6 A machine costs \$45000. Its value depreciates exponentially at a rate of $x\%$ per year. After 4 years, the value of the machine is \$26000.

(a) Calculate the value of x.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The machine will be replaced when its value first falls below \$10000.
Find the number of complete years before the machine is replaced.

..... years **[3]**

7 Consider the depreciation of two different vehicles over time.

(a) A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year.
Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

(b) A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

8 The mass, M grams, of a radioactive substance decays exponentially at a rate of $r\%$ per day.
After 2 days, the mass is 80 grams.
After 5 days, the mass is 40.96 grams.

(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Find the initial mass of the substance.

..... grams **[2]**

(c) Calculate the number of complete days from the start when the mass is first less than 10 grams.

..... days **[3]**

9 The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

(a) Calculate the percentage increase in the population per week.

..... % **[4]**

(b) Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

(c) Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

1 **(a)** **(a)** $8000(1+\frac{p}{100})^3 = 27000$ **[M1]**

$1+\frac{p}{100} = \sqrt[3]{\frac{27000}{8000}} = \frac{3}{2}$ **[M1]**

50 **[A1]**

(b) $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 $450 \times \left(1+\frac{6.5}{100}\right)^{12}$ **[M1]**

450×1.065^{12} **[M1]**

958.01 (accept 958) **[A1]**

[Back to Question] *[Total Marks: 3]*

3 $32000 \times (1-0.12)^5$ oe **[M2]**

For 32000×0.88^n where $n > 1$, award **[M1]**

16887 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** $65000 \times \left(1+\frac{2.8}{100}\right)^5$ or 65000×1.028^5 **[M1]**

74624 **[A1]**

(b) $\frac{74624-65000}{65000} \times 100$ or $(1.028^5 - 1) \times 100$ **[M1]**

14.8 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** **(a)** $85000(1+\frac{r}{100})^{12} = 112000$ **[M1]**

$1+\frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

(b) $85000 \times \left(1+\frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1+\frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $45000 \times \left(1 - \frac{x}{100}\right)^4 = 26000$ **[M1]**

$1 - \frac{x}{100} = \sqrt[4]{\frac{26000}{45000}}$ or $1 - \frac{x}{100} = 0.8711...$ **[M1]**

12.9 **[A1]**

(b) $45000 \times (0.8711...)^n < 10000$ or $(0.8711...)^n < \frac{2}{9}$ **[M1]**

Testing values or using logarithms: $n = 10 \implies \$11342$ and $n = 11 \implies \$9881$ **[M1]**

11 **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** $28\,500 \times \left(1 - \frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

(b) $35\,000 \times \left(1 - \frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1 - \frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1 - \frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Let $k = 1 - \frac{r}{100}$. Then $M_0k^2 = 80$ and $M_0k^5 = 40.96$ **[M1]**

$k^3 = \frac{40.96}{80} = 0.512 \implies k = \sqrt[3]{0.512} = 0.8$ **[M1]**

$1 - \frac{r}{100} = 0.8 \implies r = 20$ **[A1]**

(b) $M_0 \times 0.8^2 = 80$ or $M_0 \times 0.64 = 80$ **[M1]**

125 **[A1]**

(c) $125 \times 0.8^n < 10$ or $0.8^n < 0.08$ **[M1]**

Testing values: $n = 11 \implies 0.8^{11} = 0.0858...$ and $n = 12 \implies 0.8^{12} = 0.0687...$ **[M1]**

12 **[A1]**

[Back to Question] *[Total Marks: 8]*

9 **(a)** $2500 \times \left(1 + \frac{r}{100}\right)^6 = 9100$ **[M1]**

$\left(1 + \frac{r}{100}\right)^6 = 3.64$ **[M1]**

$1 + \frac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept $24.03\ldots$) **[A1]**

(b) 2500×1.2403^{10} or 9100×1.2403^{4} **[M1]**

21 500 **[A1]**

(c) $2500 \times 1.2403^{n} > 50\,000 \implies 1.2403^{n} > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

[Back to Question] *[Total Marks: 9]*